

Classical Communication in BB84 Monogamy Games: Perfect Recovery and One-Message Bounds

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Original manuscript September 21, 2026; v0.4-P2 public release October 1, 2026

Public preprint release authorized by the author on October 1, 2026.

Abstract. We study the classical communication required for two separated holders of quantum side information to recover the same complete BB84 measurement outcome after the basis string is publicly revealed. Arbitrary finite-dimensional prior entanglement is allowed; post-split quantum transmission is excluded. We establish the exact perfect-recovery cost $c_{\min}(n) = n$ for finite-round classical interaction, together with a bounded-error converse. For one message with at most M values, a message-block projector bound and a coordinate-relay construction give, for integer $0 \leq c \leq n$,

$$Q_{\text{BB84}}^{n-c} \leq p_{\text{1msg}}^*(n, c) \leq \frac{1 + 2^{(c-n)/2}}{2}, \quad Q_{\text{BB84}} = \cos^2(\pi/8).$$

The bounds meet at $c = n - 1$, yielding the exact family $p_{\text{1msg}}^*(n, n - 1) = Q_{\text{BB84}}$ for every $n \geq 1$, including $p_{\text{1msg}}^*(2, 1) = Q_{\text{BB84}}$. Under the stated worst-case bit-cost convention, the interactive case $(n, c) = (2, 1)$ has the same value. The general interior identity $p_{\text{1msg}}^*(n, c) = Q_{\text{BB84}}^{n-c}$, the interactive near-full problem for $n \geq 3$, and the exact physical value at $(3, 1)$ remain open.

I. Introduction

Monogamy-of-entanglement games formalize a basic constraint of quantum information: two separated systems cannot both hold unrestricted predictive power about outcomes of incompatible measurements on a third system. The BB84 monogamy game of Tomamichel, Fehr, Kaniewski, and Wehner (TFKW) gives an especially clean example. Without post-question communication, the optimal probability that two separated receivers both recover the full n -bit BB84 outcome string is

$$\left(\frac{1}{2} + \frac{1}{2\sqrt{2}}\right)^n, \quad (1)$$

and the game obeys strong parallel repetition [1]. We write $Q_{\text{BB84}} = \cos^2(\pi/8) = 1/2 + 1/(2\sqrt{2})$.

TFKW prohibit communication after state preparation. Here we explicitly permit classical messages after the public basis reveal and charge them to a worst-case communication budget. After the referee reveals the basis string, how much classical communication is required for both receivers to recover the same complete outcome string? Under a one-message budget, how much can c classical bits improve the full-string success probability? These questions are distinct from the classical communication required to simulate Bell correlations [2, 3]; the target here is simultaneous recovery of a referee's measurement outcome by two holders of quantum side information.

Our analysis combines quantum-memory uncertainty relations [4–6], explicit classical communication accounting,

and projector bounds. The role of the TFKW method is stated in Section VI.

A. Results

The paper separates the exact cost of perfect recovery from the success probability achievable with a limited one-message budget. The main results are as follows.

1. *Operational formulation.* Section II specifies simultaneous full-string recovery, free public basis information, finite-dimensional memories, and charged classical communication. One-message and interactive values are distinguished throughout.
2. *Perfect recovery and bounded error.* Proposition 1 proves $c_{\min}(n) = n$ by a matching converse and relay construction. Proposition 2 gives a bounded-error converse from the same entropy remainder; it is not asserted tight.
3. *Message-block bound.* Lemma 1 bounds the overlap for two questions at Hamming distance t . For the same sender, the message blocks satisfy

$$PRP = \sum_{m=1}^M PR_m P \preceq M 2^{-t} P. \quad (2)$$

This yields the pair bound and the bracket in Proposition 3. For opposite senders, discarding the receiver decoders gives the bound without the factor M .

4. *Exact equality cases.* Corollary 1 proves $p_{\text{1msg}}^*(n, n - 1) = Q_{\text{BB84}}$ by matching the relay lower bound to the pair upper bound. This includes $p_{\text{1msg}}^*(2, 1) = Q_{\text{BB84}}$. The stated cost convention also gives $p_{\text{int}}^*(2, 1) = Q_{\text{BB84}}$ (Sec. V.D).
5. *Open frontier.* The general interior Q_{BB84}^{n-c} equality, interactive near-full equality for $n \geq 3$, and the exact physical $(3, 1)$ value remain open (Sec. VII).

II. Operational model

Alice holds an n -qubit register A , while Bob and Charlie hold arbitrary finite-dimensional registers B and C . The initial state ρ_{ABC} is arbitrary and may be entangled; it is prepared independently of the later basis choice. All logarithms are base two, and H denotes conditional von Neumann entropy (Shannon entropy for classical registers). Alice samples

$$\Theta = (\Theta_1, \dots, \Theta_n) \in \{0, 1\}^n \quad (3)$$

uniformly and measures qubit i in the computational basis when $\Theta_i = 0$ and the Hadamard basis when $\Theta_i = 1$. Denote the outcome string by $X \in \{0, 1\}^n$. Alice then announces Θ publicly at no communication cost.

For the unrestricted interactive model, Bob and Charlie may then perform arbitrary local quantum operations and exchange a finite sequence of classical messages. At every public history, consisting of the basis string, any shared randomness, and the transmitted transcript t , the protocol fixes the next sender and a finite message alphabet $\mathcal{M}_i(\theta, t)$; any private choice of speaker, termination, or timing that carries information is part of the classical transcript rather than a free side channel. The message cost is $\ell_i(\theta, t) := \lceil \log_2 |\mathcal{M}_i(\theta, t)| \rceil$ bits, and the total worst-case cost c is the maximum sum over complete transcripts and public inputs. This alphabet-cost convention is equivalent to a standard binary interactive protocol and makes the entropy accounting below explicit. No quantum system may be transmitted after the split. At the end, Bob and Charlie output $\hat{X}_B$ and $\hat{X}_C$.

Cost convention. A singleton alphabet conveys no information and costs zero. A private choice to speak, stop, or change timing is not free: it must be represented by charged symbols in the protocol. After such encoding the speaker is fixed at each public-history node. Local operations and deterministic zero-cost steps can be composed; terminated paths may be padded with singleton messages. Expected communication cost is not the resource studied here.

For exact recovery, the requirement is

$$\Pr[\hat{X}_B = X \text{ and } \hat{X}_C = X] = 1. \quad (4)$$

For bounded error, this probability is required to be at least $1 - \epsilon$.

For the one-message model, at most one information-bearing post-basis classical message is allowed, with alphabet size at most

$$M = 2^c. \quad (5)$$

Its sender may be Bob or Charlie according to a public deterministic function of Θ . Shared classical randomness fixed before the split is allowed, but the identity or timing of the sender is not permitted to encode additional private information beyond the declared message alphabet. Conditioning on the shared randomness reduces the optimization to deterministic public-basis-dependent sender schedules. A deterministic dummy message includes the zero-communication case.

Let $p_{\text{1msg}}^*(n, c)$ and $p_{\text{int}}^*(n, c)$ denote the suprema of joint success over all allowed states, finite-dimensional memories, and protocols with budget at most c in the respective models. The perfect-recovery cost $c_{\text{min}}(n)$ minimizes the worst-case cost over states and protocols attaining success exactly one; it is not a guarantee of recoverability from every fixed initial state.

One-message normal form. Fix a basis string η and suppose Bob is the designated sender. A general sender operation can be written as an instrument $\{\mathcal{E}_m^\eta\}_{m=1}^M$ followed by a final decoding POVM $\{D_{x|m}^\eta\}_x$ on Bob's retained system. Because no further information reaches Bob after his single message, the sender's message and final guess have the same joint statistics as the POVM

$$B_{x,m}^\eta := (\mathcal{E}_m^\eta)^*(D_{x|m}^\eta), \quad \sum_{x,m} B_{x,m}^\eta = I_B. \quad (6)$$

Charlie, after receiving m , uses a conditional POVM $\{C_{x,m}^\eta\}_x$. Since the question and message alphabets are finite, a common finite-dimensional Naimark dilation lets us take the finite family of joint sender POVMs and conditional receiver POVMs to be projective on enlarged local registers without changing any winning probability. The one-message proof below uses this normal form. The same construction applies with Bob and Charlie exchanged. This reduction does not combine a multi-round interactive transcript into one sender POVM, because a sender may then receive information before making the final guess.

III. Exact communication cost for perfect recovery

The central uncertainty input is a tripartite memory relation for mutually unbiased measurements. For each fixed product-basis string θ , the BB84 product basis indexed by θ and the complementary product basis indexed by $\bar{\theta}$ are mutually unbiased in dimension 2^n . Hence

$$H(X_\theta|B) + H(X_{\bar{\theta}}|C) \geq n, \quad (7)$$

where the conditional entropies are evaluated on the corresponding classical-quantum post-measurement states.

This is the standard tripartite quantum-memory uncertainty relation for mutually unbiased bases [4–6]; see Sec. IV.E and Eq. (206) of Ref. [6].

Averaging Eq. (7) uniformly over θ and relabeling $\bar{\theta}$ in the second term gives

$$H(X|B, \Theta) + H(X|C, \Theta) \geq n. \quad (8)$$

Proposition 1 (Perfect-recovery cost).

Under the operational model above, the minimum worst-case total classical communication required for both Bob and Charlie to recover the same complete n -bit outcome string with certainty is

$$c_{\min}(n) = n. \quad (9)$$

Proof. Let T_i denote the classical transcript after the first i messages and let B_i, C_i denote the corresponding local quantum memories. Define

$$S_i := H(X|B_i, \Theta, T_i) + H(X|C_i, \Theta, T_i). \quad (10)$$

Before communication, Eq. (8) gives $S_0 \geq n$.

For shared randomness, condition on its value throughout; it can be included in both memories. First consider a round in which Bob sends on every current history. Bob's retained system and message result from a local channel, so data processing gives

$$H(X|B_i, \Theta, T_i) \geq H(X|B_{i-1}, \Theta, T_{i-1}). \quad (11)$$

For Charlie, a classical message can reduce conditional entropy by at most its conditional Shannon entropy. With $p_{\theta,t} := \Pr[\Theta = \theta, T_{i-1} = t]$,

$$I(X; M_i|C_{i-1}, \Theta, T_{i-1}) \leq H(M_i|\Theta, T_{i-1}) \quad (12)$$

$$\leq \sum_{\theta,t} p_{\theta,t} \ell_i(\theta, t) \quad (13)$$

$$= \mathbb{E}[\ell_i(\Theta, T_{i-1})]. \quad (14)$$

The first inequality uses the nonnegativity of conditional entropy of a classical register given quantum information. Subsequent local processing cannot decrease Charlie's conditional entropy. Consequently,

$$H(X|C_i, \Theta, T_i) \geq H(X|C_{i-1}, \Theta, T_{i-1}) - \mathbb{E}[\ell_i(\Theta, T_{i-1})]. \quad (15)$$

For a history-dependent speaker, apply the sender and receiver inequalities separately within each public-history branch, with Bob and Charlie exchanged where necessary, and then average. In either case,

$$S_i \geq S_{i-1} - \mathbb{E}[\ell_i(\Theta, T_{i-1})]. \quad (16)$$

Along every complete transcript the sum of the message costs is at most c , so

$$\sum_i \mathbb{E}[\ell_i(\Theta, T_{i-1})] = \mathbb{E}\left[\sum_i \ell_i(\Theta, T_{i-1})\right] \leq c. \quad (17)$$

Summing over all rounds gives

$$S_f \geq n - c. \quad (18)$$

If both receivers decode X with certainty, measurement data processing and the nonnegativity of classical-quantum conditional entropy imply that both final conditional entropies vanish. Thus $S_f = 0$, and Eq. (18) implies $c \geq n$.

For achievability, choose the initial state to contain n EPR pairs shared between Alice's register and Bob. After Alice reveals Θ , Bob measures his retained halves in the matching product basis, obtaining Alice's string X with certainty. Bob sends the n -bit string X to Charlie. Hence $c = n$ is sufficient. $\square$

IV. A bounded-error converse

Applying the standard Fano inequality [7, Sec. 2.10, p. 39, Eq. (2.140)] to the entropy remainder in Eq. (18) gives a bounded-error converse.

Proposition 2 (Fano consequence).

Suppose $0 \leq \epsilon \leq 1 - 2^{-n}$ and

$$\Pr[\hat{X}_B = X \text{ and } \hat{X}_C = X] \geq 1 - \epsilon. \quad (19)$$

Then any worst-case total classical communication budget c obeys

$$c \geq n - 2h_2(\epsilon) - 2\epsilon \log_2(2^n - 1). \quad (20)$$

Proof. Joint success at least $1 - \epsilon$ implies that each receiver has full-string error $e_B, e_C \leq \epsilon$. Let E_B indicate whether Bob's decoded string is incorrect. Given a correct guess, X is fixed; given an incorrect guess, it has at most $2^n - 1$ possible values. The entropy chain rule therefore gives Fano's bound $H(X|\hat{X}_B, \Theta, T_f) \leq h_2(e_B) + e_B \log_2(2^n - 1)$, where h_2 is binary entropy. This expression is increasing in e_B on $[0, 1 - 2^{-n}]$. Measurement data processing then gives

$$H(X|B_f, \Theta, T_f) \leq h_2(\epsilon) + \epsilon \log_2(2^n - 1), \quad (21)$$

and the same upper bound for Charlie. Combining the two upper bounds with Eq. (18) yields Eq. (20). $\square$

Equation (20) is not asserted tight. For an error allowance $\epsilon > 1 - 2^{-n}$, replace ϵ on its right-hand side by $\min\{\epsilon, 1 - 2^{-n}\}$; this uses the maximum of the Fano expression over the allowed error range and gives only a trivial bound there.

V. Single-message communication

We now restrict post-basis communication to one message. The following Hamming-distance bound resolves the message into orthogonal blocks and sums at most M blockwise estimates.

A. Hamming-distance bound with message blocks

Let $\theta, \phi \in \{0, 1\}^n$ be two product BB84 basis strings differing in exactly t coordinates. For full outcomes x, z ,

$$|\langle x_\theta | z_\phi \rangle|^2 = \begin{cases} 2^{-t}, & x_j = z_j \text{ for every unchanged } j, \\ 0, & \text{otherwise.} \end{cases} \quad (22)$$

Lemma 1 (One-message Hamming pair bound).

Under the one-message model, suppose the same designated sender transmits a message with at most M values on both questions after the basis is revealed. Let p_θ and p_ϕ denote the simultaneous full-string success probabilities on the two questions. Then

$$\frac{p_\theta + p_\phi}{2} \leq \frac{1 + \min\{1, \sqrt{M/2^t}\}}{2}. \quad (23)$$

If the designated sender is opposite on the two questions, then

$$\frac{p_\theta + p_\phi}{2} \leq \frac{1 + 2^{-t/2}}{2}. \quad (24)$$

Proof. Use the one-message normal form above. Assume first that Bob is the sender on both questions. After a common Naimark dilation, write Bob's joint message-and-guess PVM as $\{B_{x,m}^\eta\}_{x,m}$ and Charlie's conditional PVM, for message m , as $\{C_{x,m}^\eta\}_x$, where $\eta \in \{\theta, \phi\}$. Define the coarse-grained projectors

$$B_x^\eta := \sum_m B_{x,m}^\eta, \quad B_m^\eta := \sum_x B_{x,m}^\eta. \quad (25)$$

The winning projectors are

$$\Pi_\theta = \sum_{x,m} |x_\theta\rangle\langle x_\theta| \otimes B_{x,m}^\theta \otimes C_{x,m}^\theta, \quad (26)$$

$$\Pi_\phi = \sum_{z,m} |z_\phi\rangle\langle z_\phi| \otimes B_{z,m}^\phi \otimes C_{z,m}^\phi. \quad (27)$$

Keep Bob's decoder on the θ question and Charlie's decoder on the ϕ question by defining

$$P := \sum_x |x_\theta\rangle\langle x_\theta| \otimes B_x^\theta \otimes I_C, \quad (28)$$

$$R_m := \sum_z |z_\phi\rangle\langle z_\phi| \otimes B_m^\phi \otimes C_{z,m}^\phi, \quad R := \sum_{m=1}^M R_m. \quad (29)$$

These are projectors, the R_m are mutually orthogonal, and $\Pi_\theta \preceq P$ and $\Pi_\phi \preceq R$. For a fixed message value introduce the further relaxation

$$\tilde{R}_m := \sum_z |z_\phi\rangle\langle z_\phi| \otimes I_B \otimes C_{z,m}^\phi. \quad (30)$$

Since $B_m^\phi \preceq I_B$, one has $R_m \preceq \tilde{R}_m$ and therefore

$$PR_mP \preceq P\tilde{R}_mP. \quad (31)$$

Let $S := \{j : \theta_j = \phi_j\}$ be the unchanged coordinates. Expanding the sandwich gives

$$P\tilde{R}_mP = \sum_{x,z,x'} \langle x_\theta | z_\phi \rangle \langle z_\phi | x'_\phi \rangle |x_\theta\rangle\langle x'_\theta| \quad (32)$$

$$\otimes B_x^\theta B_{x'}^\theta \otimes C_{z,m}^\phi. \quad (33)$$

Because $\{B_x^\theta\}_x$ is a PVM, $B_x^\theta B_{x'}^\theta = \delta_{x,x'} B_x^\theta$. Hence

$$P\tilde{R}_mP = \sum_{x,z} |\langle x_\theta | z_\phi \rangle|^2 |x_\theta\rangle\langle x_\theta| \otimes B_x^\theta \otimes C_{z,m}^\phi. \quad (34)$$

Equation (22) vanishes unless $z|_S = x|_S$ and equals 2^{-t} on compatible pairs, so

$$P\tilde{R}_mP = 2^{-t} \sum_x |x_\theta\rangle\langle x_\theta| \otimes B_x^\theta \quad (35)$$

$$\otimes \sum_{z: z|_S = x|_S} C_{z,m}^\phi. \quad (36)$$

For each m , $\{C_{z,m}^\phi\}_z$ is a PVM, and the compatible sum is therefore bounded by I_C . Thus

$$PR_mP \preceq P\tilde{R}_mP \preceq 2^{-t}P. \quad (37)$$

Summing the M message blocks gives the extension of the TFKW sandwich

$$PRP = \sum_{m=1}^M PR_mP \preceq \frac{M}{2^t}P. \quad (38)$$

Since P and R are projectors,

$$\|PR\|^2 = \|PRP\| \leq \min\{1, M/2^t\}. \quad (39)$$

Moreover, the range inclusions implied by $\Pi_\theta \preceq P$ and $\Pi_\phi \preceq R$ give

$$\|\Pi_\theta \Pi_\phi\| \leq \|PR\| \leq \min\{1, \sqrt{M/2^t}\}. \quad (40)$$

For two projectors A, B , $\|A + B\| \leq 1 + \|AB\|$. Therefore, for any initial state ρ ,

$$\frac{p_\theta + p_\phi}{2} = \frac{1}{2} \text{Tr}[(\Pi_\theta + \Pi_\phi)\rho] \quad (41)$$

$$\leq \frac{1}{2} \|\Pi_\theta + \Pi_\phi\| \quad (42)$$

$$\leq \frac{1 + \min\{1, \sqrt{M/2^t}\}}{2}, \quad (43)$$

which proves Eq. (23).

Now suppose Bob sends on θ and Charlie sends on ϕ . Absorb each message into the corresponding sender's joint outcome and drop the receiver decoder on both questions. Here $\{C_{z,m}^\phi\}_{z,m}$ denotes Charlie's joint sender PVM, not the conditional receiver PVM from the preceding case. With

$$C_z^\phi := \sum_m C_{z,m}^\phi, \quad (44)$$

define the mixed relaxations

$$P := \sum_x |x_\theta\rangle\langle x_\theta| \otimes B_x^\theta \otimes I_C, \quad (45)$$

$$R := \sum_z |z_\phi\rangle\langle z_\phi| \otimes I_B \otimes C_z^\phi. \quad (46)$$

The same sandwich calculation gives directly

$$PRP \preceq 2^{-t}P, \quad (47)$$

with no sum over message blocks. Hence $\|\Pi_\theta \Pi_\phi\| \leq 2^{-t/2}$, and the same two-projector estimate proves Eq. (24). $\square$

When $t = n$ and $1 \leq M \leq 2^n$, the two bases are fully complementary in dimension $d = 2^n$, and Eq. (23) reduces to

$$\frac{p_\theta + p_{\bar{\theta}}}{2} \leq \frac{1 + \sqrt{M/2^n}}{2}. \quad (48)$$

B. One-message bracket and its equality cases

Proposition 3 (One-message bracket).

For integers $n \geq 1$ and $0 \leq c \leq n$,

$$Q_{\text{BB84}}^{n-c} \leq p_{\text{1msg}}^*(n, c) \leq \frac{1 + 2^{(c-n)/2}}{2}. \quad (49)$$

Proof. For the upper bound, pair each basis string θ with its bitwise complement $\bar{\theta}$. Each pair has Hamming distance $t = n$. If the same sender is designated on both questions, Eq. (48) applies with $M = 2^c$. If the sender switches, Eq. (24) gives the smaller bound $(1 + 2^{-n/2})/2$. Therefore every complementary pair obeys

$$\frac{p_\theta + p_{\bar{\theta}}}{2} \leq \frac{1 + 2^{(c-n)/2}}{2}. \quad (50)$$

Averaging over the 2^{n-1} complementary pairs proves the upper bound. Shared randomness produces convex mixtures of deterministic sender schedules and cannot improve the maximum.

For the lower bound, use c EPR pairs between Alice's first c qubits and a fixed sender, say Bob. Prepare the remaining $n-c$ Alice qubits independently in the Breidbart state

$$|\varphi\rangle = \cos(\pi/8)|0\rangle + \sin(\pi/8)|1\rangle. \quad (51)$$

After Θ is revealed, Bob measures his EPR halves in the matching bases, learning the first c outcomes perfectly, and sends those c bits in the single message. Both parties output the relayed coordinates and guess 0 on the remaining coordinates. Each unrelayed coordinate is correct with probability Q_{BB84} , independently, yielding success Q_{BB84}^{n-c} . $\square$

Corollary 1 (Near-full one-message value).

For every $n \geq 1$,

$$p_{\text{1msg}}^*(n, n-1) = Q_{\text{BB84}}. \quad (52)$$

Also $p_{\text{1msg}}^*(n, n) = 1$.

Table 1: One-message values and bounds under the stated communication contract.

Budget or case	Value or bound	Status
$c = 0$	$p_{\text{1msg}}^*(n, 0) = Q_{\text{BB84}}^n$	TFKW
$c = n - 1$	$p_{\text{1msg}}^*(n, n - 1) = Q_{\text{BB84}}$	Exact
$c = n$	$p_{\text{1msg}}^*(n, n) = 1$	Relay
$(2, 1)$	$p_{\text{1msg}}^*(2, 1) = Q_{\text{BB84}}$	Exact
$(3, 1)$	$Q_{\text{BB84}}^2 \leq p_{\text{1msg}}^*(3, 1) \leq 3/4$	Open

Proof. Set $c = n - 1$ in Eq. (49). The upper bound becomes

$$\frac{1 + 2^{-1/2}}{2} = Q_{\text{BB84}}, \quad (53)$$

and the lower bound is $Q_{\text{BB84}}^{n-(n-1)} = Q_{\text{BB84}}$. The endpoint $c = n$ is achieved by relaying all n outcomes. $\square$

C. Selected cases and open interior points

Table 1 summarizes the exact values and the open $(3, 1)$ interval under the one-message contract.

For $(n, c) = (3, 1)$, the coordinate-relay lower bound is

$$Q_{\text{BB84}}^2 \approx 0.7285533906, \quad (54)$$

while the complementary-pair upper bound is

$$\frac{3}{4} = 0.75. \quad (55)$$

Thus the exact physical optimum remains unresolved within an interval of about 2.1447 percentage points. “Physical” here means the optimum over admissible quantum states and measurements, not a laboratory observation or a relaxation value. More generally, $p_{\text{1msg}}^*(n, c) = Q_{\text{BB84}}^{n-c}$ is a candidate identity, not a theorem, away from $c = 0, n - 1, n$.

D. Interactive one-bit consequence

Remark. Under the worst-case alphabet-cost convention of Sec. II, a protocol of total cost at most one contains at most one nontrivial message, with at most two values. Compose all deterministic singleton steps with adjacent local operations. Before the nontrivial message, its sender depends only on public information; afterwards no additional information-bearing symbol can be transmitted. A privately chosen speaker, stopping event, or timing signal must itself be encoded and charged. Thus every such protocol has the one-message normal form, with a public-basis-dependent sender; shared randomness only mixes these schedules. Therefore

$$p_{\text{int}}^*(2, 1) = p_{\text{1msg}}^*(2, 1) = Q_{\text{BB84}}. \quad (56)$$

This follows from the cost convention and Corollary 1. For $n \geq 3$, the near-full budget $n - 1 \geq 2$ allows genuine two-way interaction, so this reduction does not apply. It also does not establish $p_{\text{1msg}}^*(3, 1) = Q_{\text{BB84}}^2$.

VI. Relation to prior work and scope

Uncertainty and communication accounting. The perfect-recovery converse uses tripartite uncertainty and per-message entropy accounting; the bounded-error converse adds Fano’s inequality. Renes–Boileau, Berta et al., and the review of Coles et al. provide the quantum-memory context [4–6]. Smooth-entropy uncertainty [8] concerns a different entropy formulation; this note does not derive a finite-key security theorem. Renes–Renner study compression with quantum side information at one decoder [9]; Berta–Coles–Wehner relate complementary-outcome guessing to entanglement [10]. Those operational tasks differ from requiring both holders to recover the same full string with a charged post-basis transcript.

TFKW attribution. The message-block bound extends the Hamming-distance sandwich in the proof of TFKW’s Theorem 3 [1]. At each fixed message value m , Eq. (37) uses that calculation; summing at most M orthogonal message blocks gives Eq. (38). Pairing complementary basis strings then yields the one-message bracket, and coordinate relay supplies its lower bound. The underlying projector geometry is TFKW’s; this note makes the post-basis communication budget explicit and derives the corresponding bounds and equality cases.

Extended games and changed predicates. Johnston, Mittal, Russo, and Watrous [11], and Russo’s thesis [12], study extended nonlocal and monogamy games with separated answering players. Their game formalism does not itself supply the charged post-question transcript used here. Coladangelo, Liu, and Xie [13] study XOR and related decision predicates; their results concern different winning conditions and cannot be substituted for simultaneous full-string search.

Leakage and unclonable cryptography. Broadbent–Lord’s unclonable encryption keeps the two recovery adversaries isolated after splitting [14]. Broadbent–Culf’s leaky coset-state game gives Bob’s answer to Charlie before Charlie guesses a complementary coset label [15], Sec. 3.2. This is a close operational precursor, but uses a different ensemble and different targets. Kitagawa–Nishimaki’s leakage game [16], Theorem 4.6, gives $\text{Leak}(x)$ to the splitting adversary before distribution and basis reveal; it does require both final guesses to equal x . Its distinction here is the leakage timing and channel, not absence of a full-string predicate. The comparison therefore distinguishes the ensembles, recovery targets, and timing of the classical information.

Position verification and simulation. The position-verification application in TFKW [1], Sec. V, constrains adversaries by a timed simultaneous exchange. It is not this worst-case classical bit-budget task after a common

basis reveal. Likewise, simulating entangled measurement statistics [2, 3] has a different target from recovery of an actual referee outcome at two sites.

Historical attribution and source pinpoints are recorded in Appendix A.

VII. Open questions

The coordinate-relay construction achieves Q_{BB84}^{n-c} , but the following points remain unproved in this manuscript:

1. *General one-message equality.* Whether $p_{\text{1msg}}^*(n, c) = Q_{\text{BB84}}^{n-c}$ for the remaining interior values $1 \leq c \leq n - 2$.
2. *Interactive near-full equality.* Whether $p_{\text{int}}^*(n, n - 1) = Q_{\text{BB84}}$ for $n \geq 3$. The entropy remainder $S_f \geq 1$ does not imply this guessing-probability converse. A multi-round transcript cannot be inserted as an M -valued single-sender message without an additional argument.
3. *The physical (3, 1) value.* The proven one-message interval is $Q_{\text{BB84}}^2 \leq p_{\text{1msg}}^*(3, 1) \leq 3/4$. Exact equality with Q_{BB84}^2 has not been established; a numerical relaxation or a feasible relay alone does not close it.
4. *General interactive bounds.* The full interactive identity $p_{\text{int}}^*(n, c) = Q_{\text{BB84}}^{n-c}$ and an interactive analogue of the one-message sandwich are not established. No new solver result is used; the general interactive problem remains open.

Local general measurements are already allowed. A joint quantum measurement across the separated holders would require an additional physical resource; it is not a free operation within this contract. No experimental or quantum-hardware implementation claim is made.

VIII. Conclusion

Under the stated post-basis communication model, exact simultaneous recovery of the complete BB84 outcome requires and is achieved by $c_{\min}(n) = n$ classical bits, even with arbitrary finite-dimensional prior entanglement and finite-round interaction. The bounded-error converse quantifies the remaining entropy requirement. For one message, the message-block upper bound and coordinate-relay lower bound give the stated bracket and meet exactly at $c = n - 1$. In particular, $p_{\text{1msg}}^*(n, n - 1) = Q_{\text{BB84}}$ and $p_{\text{1msg}}^*(2, 1) = Q_{\text{BB84}}$; the cost convention also gives $p_{\text{int}}^*(2, 1) = Q_{\text{BB84}}$.

The unresolved questions are the general one-message interior frontier, the interactive near-full family for $n \geq 3$, and the exact physical (3, 1) optimum. The statements proved here and these open questions are separate.

IX. Data and code availability

This work is theoretical and uses no experimental dataset. Computational checks and frozen derivation artifacts used

during manuscript preparation are retained privately by the author. The displayed claims are supported by the analytic arguments given here; no new computation or experimental dataset is introduced in this presentation revision. Supporting derivation artifacts are not part of this public release unless separately posted by the author.

AI-assistance and review disclosure

Generative AI tools assisted with literature triage, computational checks, adversarial proof review, and language editing. The author is responsible for the mathematical claims, citations, and final manuscript. No independent human expert review was obtained; the author waived that step as a release requirement. AI-assisted checks are not independent external validation.

Acknowledgments

No external funding is declared in this manuscript version.

A. Source notes

Historical attribution. TFKW credit Christandl and Schuch’s 2010 personal communication for a proof of the single-copy constant (arXiv v3, p. 2 and Ref. [15], p. 18). The underlying communication has not been recovered. This attribution is reported through TFKW; historical recovery is not a dependency of the proofs in this paper.

Fano pinpoint. The form used here is Cover–Thomas, second edition (2006), Sec. 2.10, Eq. (2.140), p. 39, following Theorem 2.10.1 on p. 38. The argument in Proposition 2 is self-contained.

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